



slender crystal

**CHRYSOCOLLA**

Orthorhombic
H $2\frac{1}{2}$ – $3\frac{1}{2}$ • SG 1.93–2.4
This blue or greenish blue hydrated silicate of copper and aluminium forms in altered copper deposits. Crystals do not occur.

**ZINNWALDITE**

Monoclinic • H $3\frac{1}{2}$ –4
SG 2.9–3.1

This brown, grey, or green mica is potassium lithium iron aluminium aluminosilicate hydroxide, with fluorine.

**VERMICULITE**

Monoclinic
H $1\frac{1}{2}$ • SG 2.4–2.7

This green or yellow clay mineral often occurs where micas have been altered. It is a hydrated silicate of magnesium, iron, and aluminium.

**GLAUCONITE**

Monoclinic • H 2 • SG 2.4–2.95

A mica, glauconite is a potassium sodium magnesium aluminium iron aluminosilicate hydroxide. It occurs in marine sedimentary rocks.

**CHRYSTILE**

Monoclinic • H $2\frac{1}{2}$ • SG 2.53

Chrysotile is a hydrous silicate of magnesium, forming fibrous, silky, white crystals in serpentinite rock. It is the most abundant of the asbestos minerals.

**BIOTITE**

Monoclinic • H $2\frac{1}{2}$ –3 • SG 3.3

Biotite, or black mica, is a potassium iron magnesium aluminosilicate hydroxide, with fluorine. It is abundant in igneous and metamorphic rocks.

tabular biotite crystal

**TALC**

Triclinic
H 1 • SG 2.58–2.83

The softest mineral, white, grey, or greenish talc is magnesium silicate hydroxide. Its many uses include toiletries, paint, and ceramics.

**ALLOPHANE**

Amorphous
H 3 • SG 2.8

A clay mineral, this aluminosilicate hydrate is an alteration product of feldspars and other minerals. It forms crusty masses.

PYROPHYLLITE

Triclinic
H 1–2 • SG 2.65–2.9

This aluminium silicate hydroxide, of variable form and colour, occurs in low-grade metamorphic rocks. It has good insulating properties.



vitreous lustre



prismatic crystals

ROCK CRYSTAL

Trigonal
H 7 • SG 2.65

Rock crystal is a transparent, colourless variety of quartz, which is widely used as an ornament or a gemstone.

**JASPER**

Trigonal • H 7 • SG 2.6

Jasper is a variety of chalcedony, or microcrystalline quartz, and is used for jewellery. It is opaque, and usually coloured red by impurities.

white quartz vein

**AGATE**

Trigonal
H 7 • SG 2.6

Forming in cavities in lavas, agate is a type of chalcedony. It is characterized by concentric colour bands caused by impurities.